

Fernando Moreno-Pino, PhD

Intelligent Systems Lab, University of Bristol



🌐 <https://fmorenopino.github.io>

🌐 <https://github.com/fmorenopino>

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📍 School of Engineering Mathematics and Technology, BS8 1UB

📍 Oxford-Man Institute, OX2 6ED

POSITIONS

- **Intelligent Systems Lab, University of Bristol** Bristol, UK
Assistant Professor in Machine Learning *Dec. 2025 - Present*
 - **Summary:** My research lies at the intersection of deep learning, probabilistic Machine Learning (ML), and quantitative finance.
- **Oxford-Man Institute, University of Oxford** Oxford, UK
Research Associate *Dec. 2025 - Present*
 - **Summary:** Collaborating on research in ML for sequential and time-series modelling, with applications to quantitative finance.
- **Oxford-Man Institute, University of Oxford** Oxford, UK
Postdoctoral Researcher *Aug. 2023 - Nov. 2025*
 - **Summary:** I led multiple first-author research projects on ML methods for spatio-temporal and sequential modeling and their application in quantitative finance, publishing in top venues in both fields.
- **Mathematical Institute, University of Oxford** Oxford, UK
Teaching Assistant *Aug. 2023 - Nov. 2025*
 - **Summary:** Teaching the deep learning course at the MSc in Mathematical and Computational Finance, University of Oxford.
- **Universidad Carlos III de Madrid** Madrid, Spain
Research & Teaching Associate, Signal Processing and Learning Group *Sep. 2018 - July 2023*
 - **Supervisor:** Prof. Dr. Antonio Artés Rodríguez.
 - **Summary:** I collaborated with Universidad Carlos III de Madrid in teaching and research activities.
- **Oxford-Man Institute, University of Oxford** Oxford, UK
Visiting Researcher. Founded by MAN Group *May 2022 - Oct. 2022*
 - **Supervisor:** Dr. Stefan Zohren.
 - **Summary:** Studying and developing of novel neural-based methods for the problems of assets' volatility forecasting and estimation of fill probabilities in Limit Order Books (LOB).
- **Universidad Carlos III de Madrid** Madrid, Spain
Research Assistant, Signal Processing and Learning Group *Dec. 2017 - Sep. 2018*
 - **Supervisor:** Prof. Dr. Antonio Artés Rodríguez.
 - **Summary:** My work focused on applying Machine Learning techniques for the Human Activity Recognition (HAR) problem.
- **Universidad de Málaga** Málaga, Spain
Research Assistant, Department of Programming Languages and Computer Science *Jan. 2016 - Sep. 2016*
 - **Supervisor:** Prof. Dr. Pedro Merino Gómez.
 - **Summary:** Research on low latency communication systems software at the MORSE Research Group.

EDUCATION

- **Universidad Carlos III de Madrid** Madrid, Spain
PhD Candidate in Probabilistic Machine Learning and Deep Learning (Cum Laude). *Sep. 2018 – May. 2023*
 - **Dissertation:** “Deep Attentive Time Series Modelling for Quantitative Finance”
 - **Advisors:** Prof. Dr. Antonio Artés Rodríguez and Dr. Pablo Martínez Olmos.
 - **Research:** My research included probabilistic machine learning methods, signal processing techniques integration into deep-learning architectures, the development of deep neural networks (as Transformer-based models) for time-series modelling and forecasting, and the application of ML techniques to quantitative finance-related problems. Previously, I worked with heterogeneous models for high dimensional data.
- **Universidad Carlos III de Madrid** Madrid, Spain
M.Sc. in Telecommunications Engineering (Electrical Engineering) *Sep. 2016 – Jul. 2018*
- **Universidad de Málaga** Málaga, Spain
B.Sc. in Telecommunications Engineering (Electrical Engineering) *Sep. 2012 – Jul. 2016*
 - **Graduated with Honors:** Best academic record of the class.

PUBLICATIONS

- **Moreno-Pino, F.***, Arroyo, Á.* , Waldon, H.* , Dong, X., Cartea, Á. (2024). Rough Transformers: Lightweight and Continuous Time Series Modelling through Signature Patching. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*, 2024.
- **Moreno-Pino, F.**, Zohren, S. (2024). DeepVol: Volatility Forecasting from High-Frequency Data with Dilated Causal Convolutions. In *Quantitative Finance*, Taylor & Francis, 2024.
- **Moreno-Pino, F.***, Arroyo, Á.* , Waldon, H.* , Dong, X., Cartea, Á. (2024). Rough Transformers: Lightweight Continuous-Time Sequence Modelling with Path Signatures. In “Next Generation of Sequence Modeling Architectures” workshop, ICML 2024.
- **Moreno-Pino, F.***, Arroyo, Á.* , Waldon, H.* , Dong, X., Cartea, Á. (2024). Rough Transformers for Continuous and Efficient Time-Series Modelling. In “Learning from Time Series for Health” workshop, ICLR 2024.
- **Moreno-Pino, F.**, Olmos, P. M., & Artés-Rodríguez, A. (2023). Deep Autoregressive Models with Spectral Attention. In *Pattern Recognition*, Elsevier, 2023.
- Jiménez Rama, Ó., **Moreno-Pino, F.**, Ramírez, D., Olmos, P.M. (2023). Interpretable Spectral Variational AutoEncoder (ISVAE) for time series clustering. *arXiv preprint arXiv:2310.11940*.
- Arroyo, Á*, Cartea, Á., **Moreno-Pino, F.*** & Zohren, S. (2023). Deep Attentive Survival Analysis in Limit Order Books: Estimating Fill Probabilities with Convolutional-Transformers. Presented at Euro Working Group on Commodities and Financial Modelling (EWGCFM) 2023 & in *Quantitative Finance*, Taylor & Francis, 2023.
- Martínez-García, M.* , **Moreno-Pino, F.***, Olmos, P. M., & Artés-Rodríguez, A. (2023). Sleep Activity Recognition and Characterization from Multi-Source Passively Sensed Data. *arXiv preprint arXiv:2301.10156*.
- **Moreno-Pino, F.**, Martínez-García, M., Olmos, P. M., & Artés-Rodríguez, A. (2022). Heterogeneous Hidden Markov Models for Sleep Activity Recognition from Multi-Source Passively Sensed Data. Accepted at ML4H 2022, collocated with NeurIPS.
- **Moreno-Pino, F.**, Sükei, E., Olmos, P. M., & Artés-Rodríguez, A. (2022). PyHHMM: A Python Library for Heterogeneous Hidden Markov Models. *arXiv preprint arXiv:2201.06968*, submitted to the Journal of Machine Learning Research, Machine Learning Open Source Software section.
- Ríos-Muñoz, G. R., **Moreno-Pino, F.**, Soto, N., M. Olmos, P. , Artés-Rodríguez, A., Fernández-Avilés, F., & Arenal, A. (2020). Hidden Markov Models for Activity Detection in Atrial Fibrillation Electrograms. In *2020 Computing in Cardiology* (pp. 1-4). IEEE.

- **Moreno-Pino, F.**, Porras-Segovia, A., López-Esteban, P., Artés, A., & Baca-García, E. (2019). Validation of Fitbit Charge 2 and Fitbit Alta HR against polysomnography for assessing sleep in adults with obstructive sleep apnea. *Journal of Clinical Sleep Medicine*, 15(11), 1645-1653.

OTHERS

- **Moreno-Pino, F.**, Artés-Rodríguez, A. (2019). Human Activity Recognition in Psychiatric Patients through Heterogeneous Hidden Markov Models. Machine Learning Summer School (MLSS), Moscow, Russia (Poster).
- **Moreno-Pino, F.**, Artés-Rodríguez, A. (2018). Sleep Activity Recognition through Hidden Markov Models. Data Science Summer School (DS3), Paris, France (Poster).

TEACHING

- **Engineering Science, University of Oxford** Oxford, UK
Tutor at the Intelligent Earth Centre for Doctoral Training Jan. 2025
 - **Deep Learning for Time Series Forecasting:** 2025.
- **Magdalen College, University of Oxford** Oxford, UK
Tutor at the “Oxford Study Abroad Programme” for Visiting BSc and MSc Students Sep. 2023 – Nov. 2025
 - **Deep Learning:** 2023 – 2025.
- **Mathematical Institute, University of Oxford** Oxford, UK
Teaching Assistant at the MSc in Mathematical and Computational Finance Sep. 2023 – Nov. 2025
 - **Deep Learning :** 2023 – 2025.
- **BBVA** Madrid, Spain
Teaching Staff, Associated with Fundación Universidad Carlos III Sep. 2021 – Present
 - **Deep Generative AI Course:** 2025 – Present.
 - **Advanced Machine Learning and Feature Engineering Course:** 2022 – Present.
 - **Intermediate Machine Learning and Feature Engineering Course:** 2022 – Present.
 - **Natural Language Processing (NLP) Course:** 2021 – Present.
- **Universidad Carlos III de Madrid** Madrid, Spain
Reader (Lecturer) & Teaching Assistant –Bachelor in Electrical Engineering Sep. 2018 – July 2023
 - **Signals and Systems:** 1st Semester 2023 (30 hours).
 - **Machine Learning II:** 1st Semester 2021 (38 hours).
 - **Bayesian Machine Learning, Modern Theory of Detection and Estimation:** 1st Semester 2018 – 2019 (61 hours).
 - **Communications Theory:** 1st Semester 2018 – 2019, 2023 (102 hours).
 - **Linear Systems:** 1st Semester 2018 (11 hours).

HONOR AND AWARDS

- **MAN Group & OMI’s Visitors Programme Fellowship (2022):** My research visit at Oxford-Man Institute during my PhD was funded by the OMI’s Visitors Programme (7200£).
- **UC3M Research Mobility Fellowship (July-October 2022):** Fellowship to conduct research at the University of Oxford (4716€).
- **FPU Research Mobility Fellowship (May-July 2022):** Fellowship to conduct research at the University of Oxford (4095€).
- **FPU Grant (2018-2022):** My PhD was funded by the Spanish Ministry of Education after a nationwide competitive selection process (60996,12€).
- **Research Fellowship, Signal Processing and Learning Group, Universidad Carlos III de Madrid (2017):** Awarded funding to support research for my M.Sc. dissertation (€5,250).
- **Talentia Scholarship by “Junta de Andalucía” (2017, declined):** Two years scholarship to complete the Master of Engineering at Cornell University’s New York campus (62092,68€).

*Denotes co-first authors with equal contributions.

- **‘Premios Extraordinarios de Fin de Estudios’ (2016):** This prize rewards the student with the best academic record, granted by Universidad de Málaga for my Bachelor studies.
- **‘Premios Ingenio’, Finalist (2016):** These prizes award the best thesis of the year on the field of Telecommunications Engineering, in the Region of Andalusia, Spain.
- **Research Fellowship, Department of Computer Science and Programming Languages, Universidad de Málaga (2016):** Awarded funding to support research for my B.Sc. dissertation (€2,074).

COMMITTEES

- **Engineering and Physical Sciences Research Council (EPSRC):** Since 2026 (Member of Peer Review College).
- **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECMLPKDD):** Since 2026 (Program Committee).
- **International Conference on AI in Finance (ICAIF):** 2023, 2025 (Program Committee).

REVIEWING

- **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECMLPKDD):** Since 2026.
- **Neural Networks:** Since 2025.
- **Neural Information Processing Systems (NeurIPS):** Since 2024 (Reviewer and workshop proposals reviewer).
- **Journal of Applied Mathematical Finance:** Since 2024.
- **Artificial Intelligence and Statistics (AISTATS):** Since 2023.
- **Artificial ACM International Conference on AI in Finance (ICAIF):** Since 2023.
- **Journal of Quantitative Finance:** Since 2023.
- **Pattern Recognition:** Since 2022.
- **AAAI Conference on Artificial Intelligence:** Since 2022.
- **IEEE Transactions on Neural Networks and Learning Systems:** Since 2021.
- **Journal of Biomedical and Health Informatics (JBHI):** 2020-2023.

TALKS

- **Practice-Oriented Artificial Intelligence Seminar, University of Bristol** Bristol, UK
Deep Attentive Sequential Modelling: Methods, Theory, and Applications Mar. 2026
- **SIAM Conference on Financial Mathematics and Engineering** Miami, US
Generative AI in Finance Jul. 2025
- **Department of Engineering Mathematics, University of Bristol** Bristol, UK
Deep Attentive Time Series Modelling for Quantitative Finance Jun. 2025
- **Quantitative Risk Management and Mathematical Finance, University of Vienna** Vienna, Austria
Neural Dynamical Systems for Time-Series Data Apr. 2025
- **Invited Talk at Citi Bank** New York, US
Rough Transformers: Lightweight and Continuous Time Series Modelling through Signature Patching Mar. 2025
- **The Alan Turing Institute & Mathematical Institute, University of Oxford** London, UK
Bridging Rough Paths and Deep Learning: New Frontiers [Video] Nov. 2024
- **Oxford-Man Institute, University of Oxford** Oxford, UK
Deep Autoregressive Models with Spectral Attention Mar. 2024
- **Signal Processing and Learning Group, Universidad Carlos III de Madrid** Madrid, Spain
Deep Attentive Time Series Modelling for Quantitative Finance Apr. 2023
- **Oxford-Man Institute, University of Oxford** Oxford, UK
DeepVol: Volatility Forecasting from High-Frequency Data with Dilated Causal Convolution Oct. 2022
- **Signal Processing and Learning Group, Universidad Carlos III de Madrid** Madrid, Spain
Deep Autoregressive Models with Spectral Attention Oct. 2021

CONFERENCES, SUMMER SCHOOLS, AND OTHERS

- **SIAM** Miami, US
Conference on Financial Mathematics and Engineering Jul. 2025
- **NeurIPS** Vancouver, Canada
Conference on Neural Information Processing Systems Dec. 2024
- **ICML** Vienna, Austria
International Conference on Machine Learning Jul. 2024
- **ICLR** Vienna, Austria
International Conference on Learning Representations May. 2024
- **AISTATS** Valencia, Spain
Artificial Intelligence and Statistics Apr. 2023
- **AI for Global Goals - University of Oxford** Oxford, United Kingdom
ML x Finance Aug. 2022
- **University of Sheffield** Sheffield, United Kingdom [Online]
The Gaussian Process Summer School Sep. 2021
- **University of Sheffield** Sheffield, United Kingdom [Online]
The Gaussian Process Summer School Sep. 2020
- **Liège Université** Liège, Belgium [Online]
Machine Learning Frontiers in Precision Medicine (MLFPM) Sep. 2020
- **ETH Zürich** Basel, Switzerland
Machine Learning Frontiers in Precision Medicine (MLFPM) Sep. 2019
- **Skoltech** Moscow, Russia
Machine Learning Summer School (MLSS) Aug. 2019 – Sep. 2019
- **École Polytechnique** Paris, France
Data Science Summer School (DS3) Jun. 2018

COURSES

- **University of California, Santa Cruz** Coursera [Online]
Bayesian Statistics: From Concept to Data Analysis, 4 weeks course Jul. 2021
- **DeepLearning.AI** Coursera [Online]
Structuring Machine Learning Projects, 3 weeks course May 2018
- **DeepLearning.AI** Coursera [Online]
Improving DNNs: Hyperparameter Tuning, Regularization and Optimization, 2 weeks course May 2018
- **Universidad Internacional Menéndez Pelayo** Barcelona, Spain
English Immersion Course Apr. 2018
- **DeepLearning.AI** Coursera [Online]
Neural Networks and Deep Learning, 4 weeks course Mar. 2018
- **Stanford University** Coursera [Online]
Machine Learning, 11 weeks course Feb. 2018
- **University of Washington** Coursera [Online]
Machine Learning: Classification, 7 weeks course Nov. 2017
- **University of Washington** Coursera [Online]
Machine Learning: Regression, 6 weeks course Oct. 2017

- **University of Washington** Coursera [Online]
Machine Learning Foundations, 6 weeks course Jul. 2017
- **Nvidia Corporation, CUDA Fellows Program & Universidad de Málaga** Málaga, Spain
Technical Training Course: Parallel Programming of the GPU with CUDA Jul. 2016 – Aug. 2016

LANGUAGES

- **Spanish:** Native language.
- **English:** Advanced, TOEFL:102/120.
- **French:** Basic.

PROJECTS

- **Heterogeneous Hidden Markov Model:** Python implementation of a HMM model capable of managing heterogeneous and missing data: <https://github.com/fmorenopino/HeterogeneousHMM>, <https://pyhmm.readthedocs.io/>.
- **VoIP calls:** C implementation of a Voice over IP calls' service (point-to-point audio conference). RTP over UDP was used: https://github.com/fmorenopino/c_calls.

PROGRAMMING SKILLS

- **Languages:** Python, Matlab, C, C++
- **Technologies:** Pytorch, Keras, Sklearn, Jupyter, Git, L^AT_EX

REFEREES

- Dr. Álvaro Cartea, University of Oxford, UK.
- Dr. Stefan Zohren, University of Oxford, UK.
- Dr. Antonio Artés Rodríguez, Universidad Carlos III de Madrid, Spain.
- Dr. Pablo Martínez Olmos, Universidad Carlos III de Madrid, Spain.